🔬 Synthetic Monitoring Fundamentals

Series: SYNTH | **Notebook:** 1 of 6 | **Created:** December 2025

Understanding Proactive Availability and Performance Testing

This notebook introduces Dynatrace Synthetic Monitoring, which enables proactive testing of application availability, functionality, and performance from locations around the world.

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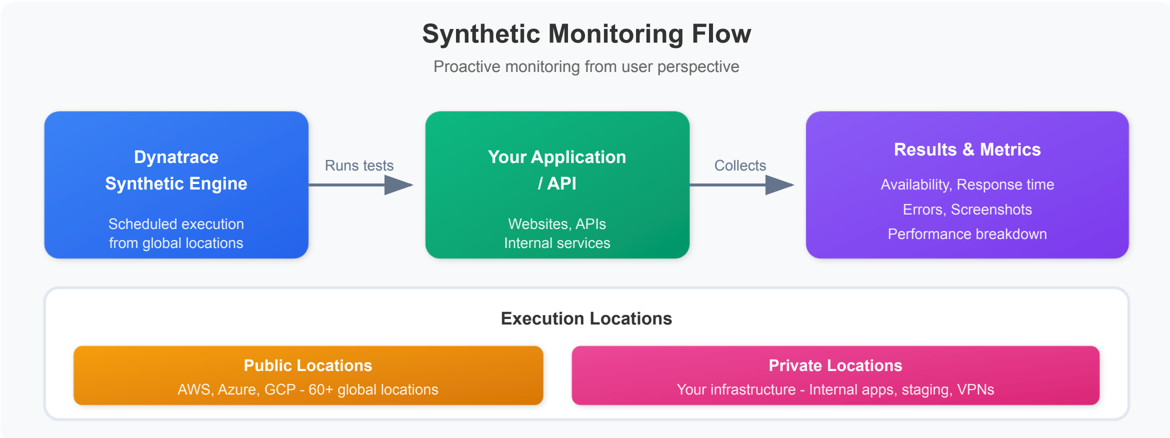
Prerequisites

- ✅ Access to a Dynatrace environment with Synthetic Monitoring enabled
- ✅ DQL query permissions (viewer role minimum)
- ✅ Basic understanding of web applications and APIs

1. What is Synthetic Monitoring?

Synthetic Monitoring simulates user interactions with your applications from external locations, providing:

- **Proactive Detection**: Find issues before real users encounter them
- **24/7 Availability Testing**: Monitor even when no users are active
- **Global Performance Baseline**: Measure response times from multiple locations
- **SLA Validation**: Verify service level agreements are being met
- **Third-Party Dependency Monitoring**: Test external APIs and services



Synthetic vs Real User Monitoring (RUM)

Aspect	Synthetic Monitoring	Real User Monitoring
Data Source	Simulated transactions	Actual user sessions
Coverage	24/7, consistent	Only when users active
Locations	Controlled, specific	Wherever users are
Use Case	Baseline, SLA, proactive	Actual experience
Cost	Per execution	Per session

2. Monitor Types

Dynatrace offers three types of synthetic monitors:

Browser Monitors (Single-URL and Browser Clickpath)

Type	Description	Use Case
Single-URL	Load a single page	Homepage availability
Browser Clickpath	Multi-step user journey	Login flows, checkout

Capabilities:
- Full browser rendering (Chrome)
- JavaScript execution
- Visual validation (screenshots)
- Performance metrics (W3C timing)
- Resource waterfall analysis

HTTP Monitors

Type	Description	Use Case
Single Request	One HTTP call	API health check
Multi-step	Chained requests	API workflow validation

Capabilities:
- Any HTTP method (GET, POST, PUT, DELETE)
- Custom headers and authentication
- Response validation (status, content, JSON)
- SSL certificate monitoring
- Lightweight and fast execution

Third-Party Monitors

Integration with external synthetic providers:
- Catchpoint
- Pingdom
- Site24x7

3. Key Concepts

Execution Locations

Location Type	Description	Best For
Public	Dynatrace-hosted worldwide	External-facing apps
Private	Your infrastructure	Internal apps, security

Execution Frequency

How often the monitor runs:
- **Browser**: 5, 10, 15, 30, 60 minutes
- **HTTP**: 1, 5, 10, 15, 30, 60 minutes

Outage Detection

Dynatrace detects outages based on:
- Consecutive failures from single location
- Failures from multiple locations simultaneously
- Local outage vs global outage classification

Key Metrics

Metric	Description	Monitor Type
`availability`	Success rate (%)	All
`responseTime`	Total execution time	All
`dnsTime`	DNS lookup duration	HTTP, Browser
`connectTime`	TCP connection time	HTTP, Browser
`sslTime`	SSL handshake time	HTTP, Browser
`ttfb`	Time to first byte	HTTP, Browser
`visuallyComplete`	Visual rendering complete	Browser
`speedIndex`	Visual progress score	Browser

4. Synthetic Data in Grail

Synthetic execution data is stored in Grail and queryable via DQL:

Data Tables

Table	Description
`dt.entity.synthetic_test`	Monitor definitions
`dt.entity.synthetic_location`	Execution locations
`dt.entity.http_check`	HTTP monitor entities
`dt.entity.browser_monitor`	Browser monitor entities

Key Fields

Field	Description
`dt.entity.synthetic_test`	Monitor entity ID
`dt.entity.synthetic_location`	Location entity ID
`synthetic.monitor.name`	Monitor display name
`synthetic.location.name`	Location display name
`synthetic.execution.id`	Unique execution identifier
`synthetic.availability`	Success (true/false)
`synthetic.response_time`	Execution duration (ms)

5. Your First Synthetic Query

Let's explore synthetic monitoring data in your environment.

```
// Discover synthetic monitors in your environment
// Note: Monitor type and enabled status are not available as entity fields
// Use bizevents to see execution details by monitor type
fetch dt.entity.synthetic_test
| fields id, entity.name
| sort entity.name asc
| limit 50
```

```
// Count monitors by event type (from execution data)
// Since entity 'type' field is not available, we count from bizevents
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| summarize {count = count()}, by: {event.type}
| sort count desc
```

```
// List available synthetic locations
// Note: cloudPlatform, city, countryCode fields are not available on
synthetic_location entity
fetch dt.entity.synthetic_location
| fields id, entity.name
| sort entity.name asc
| limit 50
```

```
// Query synthetic execution results (last 24 hours)
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| fields timestamp,
         event.type,
         synthetic_test_name = dt.entity.synthetic_test,
         location = dt.entity.synthetic_location,
         availability = toDouble(synthetic.availability),
         response_time = toDouble(synthetic.response_time)
| sort timestamp desc
| limit 100
```

```
// Synthetic availability summary (last 24 hours)
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| summarize {
    total_executions = count(),
    successful = countIf(synthetic.availability == true),
    failed = countIf(synthetic.availability == false)
  }
| fieldsAdd availability_pct = round((successful * 100.0) / total_executions,
decimals: 2)
```

```
// Response time by monitor (last 24 hours)
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter isNotNull(synthetic.response_time)
| summarize {
    avg_response_ms = avg(toDouble(synthetic.response_time)),
    max_response_ms = max(toDouble(synthetic.response_time)),
    executions = count()
  }, by: {dt.entity.synthetic_test}
| sort avg_response_ms desc
| limit 20
```

Summary

In this notebook, you learned:

✅ **What Synthetic Monitoring is** and how it differs from RUM
✅ **Monitor types** - Browser (single-URL, clickpath) and HTTP monitors
✅ **Key concepts** - Locations, frequency, outage detection
✅ **Grail data model** - Entity tables and execution fields
✅ **Basic DQL queries** - Discover monitors, locations, and results

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Next Steps

Continue to **SYNTH-02: Browser Monitors** to learn how to create and optimize browser-based synthetic tests.

References

- Synthetic Monitoring Overview
- Browser Monitors
- HTTP Monitors
- Synthetic Locations

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.